MILITARY SPECIFICATION

GUNNER'S DISPLAY FOR AN/VSG-2

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Gunner's Display for the Sight, Thermal, Tank Mounted, AN/VSG-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, and Seals
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

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MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-117	- Bags, Sleeves and Tubing-Interior Packaging
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-O-13830	- Optical Components for Fire Control Instruments; General Specification Covering the Manufacture, Assembly, and Inspection of
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible for Packaging
MIL-E-49383	- Eyepiece, Biocular, AN/VSG-2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners, or Palletized Type Base
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment
MS-24535	- Fasteners-Container Closure

DRAWINGS

DLSM-B-804912	- Gunner's Display, AN/VSG-2
SM-C-805906	- Image Intensifier, 40 Millimeter

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition. The Gunner's Display, AN/VSG-2, herein known as the gunner's display, is a major component of the Tank Mounted Thermal Sight, AN/VSG-2. The gunner's display contains an 8 power daylight telescope, a thermal channel display, a reticle projector, and the AN/VSG-2 system controls. In conjunction with the head assembly, it displays both daylight and infrared scenes to the tank gunner, displays fire control solutions generated by the fire control computer system, and allows adjustments of functions through the periscope to suit viewing requirements.

3.2 Construction. The gunner's display shall be constructed in accordance with DLSM-B-804912 and as specified herein.

3.3 First article. When specified in the contract (see 6.2), the contractor shall furnish 5 gunner's displays for first article inspection and approval.

3.4 Parts, materials and processes. Parts, materials and processes shall be as specified herein and as shown on the applicable drawings. Parts, materials and processes not specified shall be selected in accordance with MIL-P-11268.

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3.5 Major components. The gunner's display consists of the following components and subassemblies and shall be constructed in accordance with applicable drawings, specifications and as specified herein.

a. Image Intensifier, 40 Millimeter - SM-C-805906
b. Eyepiece, Biocular, AN/VSG-2 - MIL-E-49383
c. 8X Daylight Objective Assembly
d. 8X Telescope Assembly
e. Reticle Projector Assembly
f. Gunner's Display Subassembly
g. LED Prism Assembly
h. IR Boresight Assembly
i. Rear Cover Assembly
j. Image Intensifier Prism Assembly
k. Image Intensifier Power Supply Assembly
l. Built-In Test Equipment (BITE)
m. Gunner's BITE Printed Wiring Board (PWB)

3.6 Preconditioning. Each gunner's display shall be subjected to a 48-hour preconditioning prior to delivery.

3.7 Operation.

3.7.1 Power. The power requirements of the gunner's display shall not exceed the following:

+18 to +30 Volts direct current (Vdc) (Standby)	25 mA
+18 to +30 Vdc (Operate)	600 mA
+5 Vdc (BITE)	200 mA
+5 Vdc (HVPS)	45 mA
115 Vac, 400 Hz	20 mA

3.7.2 Luminance uniformity. With a uniform scene input, the luminance of each display shall not vary more than ±15 percent from the average display luminance over the central 50 percent of the display.

3.7.3 Spurious signals. Spurious images shall not be perceptible on the display. There shall be no discernible image persistence, and no discernible vibration of the raster lines. There shall be no perceptible rotation of the raster.

3.7.4 Sealing. The gunner's display shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5, ±1 psig for a minimum of 2 hours. The interior atmosphere of the units shall contain dry nitrogen (having a maximum dew point of -32°C at atmospheric pressure) at a pressure of 0.25 to 5.0 psig when delivered.

3.7.5 Zero reference for reticle projector unit (RPU). Zero reference shall be a measured voltage of 0.0000, ±0.0050 Vdc at the output of the reticle motor tachometer.

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3.7.6 Boresight excursion (8X and thermal). The reticle shall be moveable with the boresight knobs a minimum of 9 mils up, 11 mils down, 10 mils right and 10 mils left from the zero reference point specified in 3.7.5.

3.7.7 Reticle displacement accuracy and linearity. For any displacement of the reticle (left or right), the voltage shall be 0.1464 volt per mil of displacement ±0.0293 volt in the 8X daylight and 0.1464 volt per mil of displacement ±0.0200 volt in the thermal display from the zero reference point specified in 3.7.5.

3.7.8 Electromechanical backlash. The electromechanical backlash of the reticles shall not exceed 0.10 mil.

3.7.9 Reticle parallax (8X). Parallax between the image of the reticle pattern and the target image shall not exceed 0.1 mil when the target is at 1200, ±100 meters.

3.7.10 Reticle boresight backlash. Backlash in any boresight knob shall not exceed 0.1 mil.

3.7.11 Diopter zero setting. The diopter scale shall indicate zero ±0.25 diopter when an image and a reticle are focused at 0 diopter.

3.7.12 Torque, diopter. The running torque required to rotate the diopter movement through the specified range shall be within the limits of 1.5 and 7.0 inch-pounds at ambient conditions (see 6.3). When checked at the extreme temperatures of 3.8.1.1 and 3.8.2.1, the running torque shall be within the limits of 1.5 and 15 inch-pounds.

3.7.13 Boresight knob, torque. The running torque required to rotate the boresight knobs through the range specified in 3.7.6 shall not be greater than 4.0 inch-pounds when checked at ambient conditions.

3.7.14 Line of sight (LOS) variation, motor driven. Motion of the reticle in elevation shall not exceed ±0.2 mil when the reticle is traversed over the full excursion (±25.0 mils) relative to the image of a horizontal line.

3.7.15 Parallelism of reticle and image. The horizontal line of the reticle pattern through the boresight cross shall be parallel to the gunner's horizontal mounting surface within 2 degrees in both the 8X daylight and thermal channels.

3.7.16 Controls.

3.7.16.1 Contrast. Each contrast control shall vary the contrast.

3.7.16.2 Brightness. Each brightness control shall adjust the background level of the image.

3.7.16.3 Polarity. Each polarity selector shall reverse the image contrast and cause the display to represent hot or cold targets as either bright or dark.

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3.7.16.4 Mode select. When activated, the mode select shall switch between the OFF, Standby, and Operation modes.

3.7.16.5 Reticle brightness. The reticle brightness shall vary the illumination of all reticles from off to full brightness.

3.7.16.6 Boresight. The boresight controls shall adjust the azimuth or elevation in their respective channels.

3.7.16.7 Filter select. The filter select control shall select either clear, haze, red, or neutral density filter for the 8X daylight eyepiece.

3.7.16.8 Test switch. The test switch shall cause all the test lamps to illuminate.

3.7.17 Reticle total travel. The reticle shall traverse in azimuth 29, ±1 mils either side of the zero reference position as defined in 3.7.5.

3.7.18 Reticle limit switches. The reticle limit switches shall activate when the reticle is positioned at least 25.0 mils and less than 27.3 mils on either side of the zero reference position. The reticle limit switches shall activate prior to the return of the reticle past the 25 mil point.

3.7.19 Reticle motion motor. With the input voltages in phase, the reticle shall move to the right. With the input voltages 180 degrees out of phase, the reticle shall move to the left.

3.7.20 Reticle motor starting voltage. A control phase voltage of 8 volts root-mean-square (Vrms) shall drive the motor in each direction with the appropriate phase shift.

3.7.21 Reticle pattern. The reticle pattern shall be as shown in figure 1.

3.7.22 8X resolution. The 8X channel resolution shall not be greater than 6 seconds of arc.

3.7.23 Image tilt. A line which is parallel to the gunner's display/head assembly mounting surface shall appear parallel in the eyepieces of the gunner's display parallel to the mounting surface within 2 degrees.

3.7.24 Diopter backlash. Backlash in the diopter movement shall not exceed 0.250 diopter as read on the diopter scale.

3.8 Environment.

3.8.1 Low temperature.

3.8.1.1 Operation. The gunner's display shall not be damaged when operated at temperatures to -32°C (see 6.3).

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3.8.1.2 Storage. The gunner's display shall not be damaged when stored at temperatures to -57°C.

3.8.2 High temperature.

3.8.2.1 Operation. The gunner's display shall not be damaged when operated at temperatures to +52°C.

3.8.2.2 Storage. The gunner's display shall not be damaged when stored at temperatures to +71°C.

3.8.3 Temperature shock. The gunner's display shall not be damaged when subjected to sudden changes in temperature between -57°C and +71°C following a 4-hour soak at each temperature.

3.8.4 Altitude.

3.8.4.1 Operation. The gunner's display shall not be damaged when operated at altitudes up to 10,000 feet above sea level.

3.8.4.2 Transportation. The gunner's display shall not be damaged when transported at altitudes up to 40,000 feet above sea level.

3.8.5 Shock.

3.8.5.1 Basic shock. The gunner's display shall not be damaged when subjected to basic shock of 28 g's at 6 milliseconds duration.

3.8.5.2 High intensity shock. The gunner's display shall not be damaged when subjected to high intensity shock of 65 g's at 0.75 milliseconds duration.

3.8.5.3 Bench handling shock. The gunner's display shall not be damaged when subjected to shocks encountered in bench handling.

3.8.6 Vibration. The gunner's display assembly shall not be damaged when subjected to the following random and superimposed sinusoidal levels for a period of 15 minutes in each of three mutually perpendicular axes (group C: 5 minutes in each axis).

Random:	1×10^{-3} g^2/Hz, 50 to 500 Hz
Sinusoidal:	0.26 g's, X axis 0.78 g's, Y axis 0.85 g's, Z axis
	Octave 42 to 84 Hz

3.8.7 Waterproofness. The gunner's display shall withstand water hosing at a minimum of 5 pounds per square inch (psig) through a 1-inch nozzle parallel to the joints from a distance of 3 feet maximum for a time of 5 minutes minimum, without intake of water or damage.

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3.8.8 Humidity. The gunner's display shall not be damaged by exposure to 100 percent relative humidity.

3.8.9 Dust. The gunner's display shall not be damaged when exposed to dust at velocities up to 1750, ±250 feet per minute (fpm) for 12 hours.

3.9 Reliability. Not applicable.

3.10 Workmanship. Workmanship shall be as specified in MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Materials inspection. The supplier is responsible for processes, parts, and materials used in manufacturing the gunner's display in accordance with referenced specifications, standards, and drawings as specified herein.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)
c. Inspection of packaging (4.10)

4.3 First article inspection. When specified in the contract or purchase order, the contractor shall furnish 5 gunner's displays for first article inspection (see 6.2).

4.3.1 Inspection. Each assembled first article gunner's display shall be examined in accordance with table I. Presence of one or more defects shall be cause for rejection of the gunner's display.

4.3.2 Preconditioning. Each gunner's display which has passed the inspection of 4.3.1 will be subjected to the preconditioning of 4.6.

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TABLE I. Inspection.

Defects	Requirement paragraph	Inspection criteria
101. Any part or component missing or damaged	3.2	MIL-STD-252
102. Finishes not protecting substrates (scratches, etc)	3.2	MIL-STD-252
103. Scratched or chipped optics	3.2	MIL-STD-252
104. Improper or absent marking or labels	3.2	MIL-STD-252
105. Color of finishes incorrect	3.2	MIL-STD-252
106. Workmanship not as specified	3.10	MIL-STD-252
107. Controls	3.7.16	4.7.16

4.3.3 Tests.

4.3.3.1 First article inspection. From gunner's displays that have passed the preconditioning of 4.3.2, 5 first article gunner's displays shall be subjected to all tests of table II, in any order. Unless otherwise specified, the tests of table II shall be performed with the gunner's display at the ambient conditions (see 6.3). Failure of any gunner's display to pass all tests in table II shall constitute failure of first article.

TABLE II. First article inspection.

Inspection	Requirement Paragraph	Test Paragraph
Power	3.7.1	4.7.1
Luminance uniformity	3.7.2	4.7.2
Spurious signals	3.7.3	4.7.3
Sealing	3.7.4	4.7.4
Zero reference for reticle projector unit (RPU)	3.7.5	4.7.5
Boresight excursion (8X and thermal)	3.7.6	4.7.6
Reticle displacement accuracy and linearity	3.7.7	4.7.7
Electromechanical backlash	3.7.8	4.7.8
Reticle parallax (8X)	3.7.9	4.7.9
Reticle boresight backlash	3.7.10	4.7.10
Diopter zero setting	3.7.11	4.7.11
Torque, diopter	3.7.12	4.7.12
Boresight knob torque	3.7.13	4.7.13
Line of sight (LOS) variation, motor driven	3.7.14	4.7.14

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TABLE II. First article inspection. - Continued

Inspection	Requirement Paragraph	Test Paragraph
Parallelism of reticle and image	3.7.15	4.7.15
Controls	3.7.16	4.7.16
Reticle total travel	3.7.17	4.7.17
Reticle limit switches	3.7.18	4.7.18
Reticle motion motor	3.7.19	4.7.19
Reticle motor starting voltage	3.7.20	4.7.20
Reticle pattern	3.7.21	4.7.21
8X resolution	3.7.22	4.7.22
Image tilt	3.7.23	4.7.23
Diopter backlash	3.7.24	4.7.24
Low temperature	3.8.1	4.8.1
High temperature	3.8.2	4.8.2
Temperature shock	3.8.3	4.8.3
Altitude	3.8.4	4.8.4
Shock	3.8.5	4.8.5
Vibration	3.8.6	4.8.6
Waterproofness	3.8.7	4.8.7
Humidity	3.8.8	4.8.8
Dust	3.8.9	4.8.9

4.4 Quality conformance inspection.

4.4.1 Inspection. Each assembled gunner's display shall be examined in accordance with table I. Presence of one or more defects shall constitute failure of that gunner's display.

4.4.2 Preconditioning. Each gunner's display which has passed the inspection of 4.4.1 shall be subjected to the preconditioning of 4.6.

4.4.3 Group A inspection. Each gunner's display that has passed the preconditioning of 4.4.2 shall be inspected for the attributes specified in table III in any sequence. Failure of any gunner's display to pass all the tests of table III shall constitute failure of that gunner's display.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Zero reference for reticle projector unit (RPU)	3.7.5	4.7.5
Boresight excursion (8X and thermal)	3.7.6	4.7.6
Reticle displacement accuracy	3.7.7	4.7.7
Electromechanical backlash	3.7.8	4.7.8
Line of sight (LOS) variation	3.7.14	4.7.14

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TABLE III. Group A inspection. - Continued

Inspection	Requirement paragraph	Test paragraph
Parallelism of reticle and image	3.7.15	4.7.15
Controls	3.7.16	4.7.16
Reticle total travel	3.7.17	4.7.17
Reticle limit switches	3.7.18	4.7.18
Reticle motion motor	3.7.19	4.7.19
Image tilt	3.7.23	4.7.23

4.4.3.1 Group B inspection. Not applicable.

4.4.4 Group C inspection. Inspection lots (see 6.2) shall be formed from gunner's displays which have passed the Group A inspection of 4.4.3.

4.4.4.1 Monthly Group C. Five gunner's displays shall randomly be selected from each lot and inspected for the attributes specified in table IV. Failure of any gunner's display to pass all the tests of table IV shall constitute Group C failure.

TABLE IV. Group C monthly inspection.

Inspection	Requirement paragraph	Test paragraph
Power	3.7.1	4.7.1
Luminance uniformity	3.7.2	4.7.2
Spurious signals	3.7.3	4.7.3
Sealing	3.7.4	4.7.4
Reticle parallax (8X)	3.7.9	4.7.9
Reticle boresight backlash	3.7.10	4.7.10
Diopter zero setting	3.7.11	4.7.11
Torque, diopter	3.7.12	4.7.12
Boresight knob torque	3.7.13	4.7.13
Reticle motor starting voltage	3.7.20	4.7.20
Reticle pattern	3.7.21	4.7.21
8X resolution	3.7.22	4.7.22
Diopter backlash	3.7.24	4.7.24

4.4.4.2 Tri-monthly Group C. Two gunner's displays which have passed the tests of 4.4.3 and 4.4.4.1 shall randomly be selected from the first lot and every third lot thereafter. The two gunner's displays shall be tested per table V in any order. Following completion of the tests of table V, the tests of tables III and IV shall be re-performed. Failure of any gunner's display to pass all tests of table V, table III, and table IV shall constitute Group C failure.

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TABLE V. Group C tri-monthly inspection.

Inspection	Unit	Requirement paragraph	Test paragraph
Low temperature	2	3.8.1	4.8.1
High temperature	2	3.8.2	4.8.2
Temperature shock	2	3.8.3	4.8.3
Altitude	1	3.8.4	4.8.4
Shock	1	3.8.5	4.8.5
Vibration	1	3.8.6	4.8.6
Waterproofness	1	3.8.7	4.8.7
Humidity	1	3.8.8	4.8.8

4.4.4.3 Group C failure criteria. Actions relative to Group C failure shall be as specified in the contract (see 6.2).

4.4.4.4 Reinspection of conforming Group C gunner's display. Gunner's displays which have been subjected to and passed Group C inspection of 4.4.4.1 shall be accepted on contract or order. Gunner's displays which have been subjected to and passed Group C inspection of 4.4.4.2 shall have all damage (see 6.3) repaired and be reinspected per 4.4.3. Those gunner's displays which subsequently pass all tests of 4.4.3 shall be accepted on contract.

4.5 Group D inspection. Not applicable.

4.6 Preconditioning. Prior to conducting any tests, each gunner's display shall be subjected to a minimum of 6 preconditioning cycles. Each cycle shall conform to the profile of figure 2. The temperature extremes shall be $-32^{\circ}C$ and $+52^{\circ}C$. The unit shall be vibrated at 2.2 g's at a nonresonant frequency between 20 and 60 Hz for 10 minutes each operating hour. If damage occurs, the unit shall be removed from this test, repaired, and returned to preconditioning. The preconditioning shall continue for at least one full cycle after repair. Failure to successfully complete preconditioning shall constitute failure of this test.

4.7 Tests. All tolerances herein are absolute. Any test measuring equipment tolerances shall be subtracted from the gunner's display tolerance. Unless specified no internal adjustments or component gunner's display replacement shall be made on any gunner's display subsequent to the start of any of the tests specified herein.

4.7.1 Power. The gunner's display shall be operated in both the standby and operate modes and each current and voltage requirement measured. Failure to meet the requirements of 3.7.1 shall constitute failure of this test.

4.7.2 Luminance uniformity. The line-to-line uniformity of the thermal channel shall be measured by using a photometer to scan the raster in the vertical direction in the center, right and left half of the biocular. The photometer used shall have an aperture such that its effective field of view

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is less than the width of one line in the raster. Record the photometer output using an x-y recorder. Luminance variance in excess of ±15 percent from the average display luminance over the central 50 percent of the display shall constitute failure of this test.

4.7.3 Spurious signals. The periscope entrance shall be covered and the system adjusted for maximum brightness and midrange contrast. The thermal channel display shall be observed. Any discernible spurious images, image persistence, vibration of the raster lines or raster rotation shall constitute failure of this test.

4.7.4 Sealing. Before the start of the sealing test, all plates, covers, access doors, etc., shall be removed and installed two times. The sealing test shall be performed using a pressure gage capable of reading between 0.25 and 6.0 pounds per square inch (psig) with a resolution of 0.2 psig or less. Pressurize the unit with dry nitrogen gas having a dew point at least as low as -32°C to 5.0, ±1.0 psig. Retain the applied pressure in the unit for a minimum of 2 hours. All knobs, switches, levers, actuators, and controls shall be moved to their extreme positions at least three times during the 2-hour period. At the end of the 2-hour period, there shall be no evidence of leakage in excess of 0.2 psig as indicated on the pressure gage. Upon satisfactory completion of the sealing test, the units shall be flushed with the dry nitrogen gas specified herein. The interior atmosphere of 10 units shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as -32°C. When this atmosphere is attained, the unit shall be sealed with a delivery pressure as specified in 3.7.4. Failure to meet the requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Zero reference for reticle projector unit (RPU). Apply 5 Vrms, phase A, 400 Hz between pins A and B of connector 2J2 of the gunner's display and 5 Vrms, phase A inverted, 400 Hz between pins J and L of connector 2J2, measure the voltage of pin M with respect to pin L of connector 2J2. Position the reticle traversing mechanism until a reading of 0.0000, ±0.0050 volt is obtained. Failure to obtain a reading of 0.0000, ±0.0050 volt shall constitute failure of this test.

4.7.6 Boresight excursion (8X and thermal). With the reticle positioned in zero reference (see 4.7.5) and in coincidence with the image of an incident line of sight parallel to the gunners/tank mounting surface and perpendicular to the azimuth reference of the mounting surface, the reticle shall move a minimum of 10 mils right, 9 mils up, 10 mils left, and 9 mils down. Using a measuring telescope, view the reticle through first the 8X and then the biocular eyepiece, adjust the reticle with the boresight knobs and record the amount of movement in each direction from the starting point. Failure of either reticle to move the required amount in each direction (right, left, up and down) shall constitute failure of this test.

4.7.7 Reticle displacement accuracy. With 5 volts Rms, phase A, 400 Hz applied between pins A and B of connector 2J2 of the gunner's display and 5 Vrms, phase A inverted, 400 Hz applied between pins J and L of connector 2J2, for any displacement of the reticle (left or right), the voltage between

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pins M and L of 2J2 shall be 0.1464 volt per mil of displacement ±0.02930 volt in the 8X daylight eyepiece, and 0.1464 volt per mil of displacement ±0.0200 volt in the thermal display from the center of its motion as defined by 3.7.5. Failure to meet the requirements of 3.7.7 shall constitute failure of this test.

4.7.8 Electromechanical backlash. With the reticle centered on the target and the reticle zeroed as in 4.7.5, displace the reticle to the right at least 5 mils and return to zero. If the zero position is passed, return to 5 mils right and approach zero again. With the reticle recentered on the target, the voltage between pins M and L of 2J2 shall be 0.000, ±0.015 volt. Failure to meet the requirements of 3.7.8 shall constitute failure of this test.

4.7.9 Reticle parallax (8X). View a crosshair target focused 1200, ±100 meters from the periscope 8X eyepiece. Adjust the 8X diopter scale to zero. Superimpose the 8X reticle on the 1200 meter target. While viewing through the eyepiece, parallax shall be recognized by any apparent displacement of the intersecting point of the target crosshairs with respect to the 8X reticle center dot when the observer's head is moved. Any displacement between the two images exceeding 0.1 mil shall constitute failure of this test.

4.7.10 Reticle boresight backlash. This test shall be performed on both the 8X and thermal reticles. The system reticle shall be positioned on a target that represents the 0.1 mil and backlash boundaries. The position of the system reticle shall be noted from the 7 mil scale located on the knobs. Each knob shall be individually moved 1 mil and then back to its original position. After the knob has been returned to its original position, the reticle shall be coincident with the telescope reticle. Backlash exceeding 0.1 mil except on the unity channel shall constitute failure of this test.

4.7.11 Diopter zero setting. Adjust the 8X diopter scale to 0.0. View a target or the reticle through a dioptometer and measure the actual focus position. Failure of the dioptometer to read 0.00, ±0.25 diopter shall constitute failure of this test.

4.7.12 Torque, diopter. Using a torque meter/scale, measure the running torque required to rotate the diopter movement through the specified range. A measured running torque outside the limits of 1.5 and 7.0 inch-pounds at ambient conditions and 1.5 and 15 inch-pounds at the extreme temperatures of 3.8.1.1 and 3.8.2.1 shall constitute failure of this test.

4.7.13 Boresight knob torque. Using a torque meter/scale, measure the running torque required to move the boresight knob through the range specified in 3.7.6. A measured running torque exceeding 4.0 inch-pounds shall constitute failure of this test.

4.7.14 LOS variation, motor driven.. With the gunner's display mounted on a rotary table a reflected collimator shall be used to present, simultaneously, a point source target to the 8X daylight and thermal channels. The reticles in the 8X and thermal channels shall be positioned such that their centers are coincident with the target presented and the reticle traversing mechanism is at its zero reference position as per 3.7.5. The gunner's display shall be

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rotated 25 mils to the left using the rotary table. The reticles in the 8X and thermal channels shall be electrically driven until the reticles are again coincident in azimuth with the target source, and this movement shall be observed through a measuring telescope with two moveable filars. The filars will be positioned 0.2 mil above and 0.2 mil below the original coincident point. As the reticles are electrically driven 25 mils to the left, the center point of the reticles shall be observed to see that the filar boundaries are not exceeded. The test shall be repeated for a movement of 25 mils to the right. Any variation of the reticles in a vertical direction greater than 0.2 mil shall constitute failure of this test.

4.7.15 Parallelism of reticle and image. The horizontal line of a telescope and the target are positioned parallel to the mounting surface before the sight is mounted. The target is then viewed through both the 8X and thermal channels with the telescope. Failure of the horizontal lines of the target, the telescope, and the reticle to appear parallel within $\pm$2 degrees in both channels shall constitute failure of this test.

4.7.16 Controls. Each control shall be tested for proper operation and conformance as specified in 3.7.16. Failure of each control to function as specified in 3.7.16 shall constitute failure of this test. All controls shall be tested to ensure that a clockwise motion of the control causes an increase in the control parameter or a movement up or to the right of the controlled item and that a counterclockwise motion of the control causes a decrease in the control parameter or a movement down or to the left of the controlled item. Failure to meet the requirements of 3.7.16 shall constitute failure of this test.

4.7.17 Reticle total travel. The reticle shall be driven until the voltage between pins M and L of 2J2 equals 0.0 volt. The center of a target shall then be placed at the center of the reticle. The reticle shall be moved to the extreme limits and the amount of travel measured. Failure to meet the requirements of 3.7.17 shall constitute failure of this test.

4.7.18 Reticle limit switches. Activation of the limit switches shall be determined by monitoring pins H and L of connector 2J2 to determine when they are shorted together while the reticle is moved in azimuth. Failure to meet the requirements of 3.7.18 shall constitute failure of this test.

4.7.19 Reticle motion, motor. Apply 115 volts, 400 Hz, phase A (fixed phase) between pins A and B of 2J2, and 48 volts, 400 Hz (control phase) between pins C and F of CJ2. With the control phase in phase with the fixed phase observe the motion of the reticle. With the control phase 180° out of phase with the fixed phase, observe the motion of the reticle. Failure to meet the requirements of 3.7.19 shall constitute failure of this test.

4.7.20 Reticle motor starting voltage. With the fixed phase energized as in 4.7.19, apply a control phase voltage of 8 Vrms first in phase then 180 degrees out of phase with the fixed phase and observe the direction of motion under each condition. Failure to meet the requirements of 3.7.20 shall constitute failure of this test.

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4.7.21 Reticle pattern. The 8X and thermal channel reticles shall be visually compared with figure 1. Failure to meet the requirements of 3.7.21 shall constitute failure of this test.

4.7.22 8X resolution. This test shall be conducted utilizing an observation telescope of at least 20 power, a resolving power chart representing the angular subtense for the resolution specified in 3.7.22 and a supporting medium capable of receiving the observation telescope. Construction and positioning of the resolving power chart shall comply with the "Resolution Test" specified in MIL-O-13830. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Failure to meet the requirements of 3.7.22 shall constitute failure of this test.

4.7.23 Image tilt. A vertical target shall be placed parallel to the vertical input plane of the 8X input port, and the vertical crosshair of a telescope shall be located parallel to the vertical output plane of the 8X channel. The vertical lines of the target and telescope shall be compared and any deviations measured. Failure to meet the requirements of 3.7.23 shall constitute failure of this test.

4.7.24 Diopter backlash. The diopter shall be examined to ensure that backlash does not exceed the requirements. Failure to meet the requirements of 3.7.24 shall constitute failure of this test.

4.8 Environmental tests. Unless otherwise specified, environmental tests shall be performed in accordance with MIL-STD-810, except where modified by this specification. The performance tests required after the completion of the individual tests of MIL-STD-810 shall be as per table III. When MIL-STD-810 requires a performance test during an environmental test, the tests shall be 4.7.10, 4.7.17, and 4.7.22.

4.8.1 Low temperature. The gunner's display shall be tested in accordance with MIL-STD-810, Method 502, Procedure I, except the storage period (Step 2) shall be 12 hours at -57°C, and the operating temperature (Step 4) shall be -32°C. Failure to meet the requirements of 3.8.1 shall constitute failure of this test.

4.8.2 High temperature. The gunner's display shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The operating temperature (Step 7) shall be +52°C. Failure to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Temperature shock. The gunner's display shall be tested in accordance with MIL-STD-810, Method 503, Procedure I. Failure to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Altitude. The gunner's display shall be tested in accordance with MIL-STD-810, Method 500, Procedure I, except the chamber pressure, operating, shall be 523.0 mm Hg or 10,000 feet above sea level. The altitude, non-

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operating, shall be 40,000 feet and shall be maintained for a period not less than one hour, and shall be performed prior to the operating altitude test. Prior to operating tests, the gunner's display shall be purged with dry nitrogen. Failure to meet requirements of 3.8.4 shall constitute failure of this test.

4.8.5 Shock. The gunner's display shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure I, IV and V except testing shall be done non-operational and at ambient temperature. Figure 516.1-2 shall apply to Procedures I and IV. The parameters for Procedure I shall be 28 g's peak value with 6 ms duration. The parameters for Procedure IV shall be 65 g's peak value with 0.75 ms duration applied by 3 shocks along each axis (18 shocks total). Failure to meet the requirements of 3.8.5 shall constitute failure of this test.

4.8.6 Vibration. The gunner's display shall be tested in accordance with MIL-STD-810, Method 514.1 and procedures as modified herein. Mount the gunner's display on the vibration table. With the gunner's display non-operating, vibrate in each of three mutually perpendicular axes to the random and superimposed sinusoidal levels of 3.8.6 for 15 minutes. (Group C: 5 minutes in each axis.) The appropriate periodic component will be swept up each octave in one minute and down the octave the next minute. Failure to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Waterproofness. Subject the gunner's display to a water pressure of 5 psig through a 1-inch nozzle from a distance of 3 feet maximum in a direction parallel to the joints for 5 minutes. Wipe the system dry, remove the covers and inspect for water inside. Verify that the system is operable. Inability to meet the requirements of 3.8.7 or/accumulation of water inside the gunner's display shall constitute failure of this test.

4.8.8 Humidity. The test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure I. Failure to meet requirements of 3.8.8 shall constitute failure of this test. Failure shall be as defined in MIL-STD-810.

4.8.9 Dust. The test shall be conducted in accordance with MIL-STD-810, Method 510, Procedure I. Failure to meet requirements of 3.8.9 shall constitute failure of this test. Failure shall be as defined in MIL-STD-810.

4.9 Reliability. Not applicable.

4.10 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Cleaning. Gunner's Display, AN/VSG-2 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Gunner's Display, AN/VSG-2 shall be dried in accordance with process C-1 of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Gunner's Display, AN/VSG-2. Each Gunner's Display, AN/VSG-2, shall be individually packed method III as follows: Place each unit within a barrier bag fabricated of material conforming to MIL-B-117, type I, class B, style 2. Closure may be by staples, tape, adhesive or heat seal. Cushion each unit on all surfaces with material conforming to MIL-P-26514, type I, class 1, designed to protect all projections and absorb the shock of impact in handling and transit. Place each cushioned item into a close-fitting box conforming to PPP-B-601, overseas type. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the cover of the box. To assure reusability of the box and to provide easy removal of the cover, fasteners-container closure conforming to MS-24535 shall be utilized. Strapping shall be in accordance with the appendix of the box specification except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188 (see 6.2).

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. Each Gunner's Display, AN/VSG-2, unit packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except the container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71,

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type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. Each Gunner's Display, AN/VSG-2, unit packed as specified in 5.1.1, shall require no further packing. The unit container shall serve as the shipping container, except that the box shall be domestic type.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Gunner's Display, AN/VSG-2, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or W3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188 (see 6.2).

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129 and as follows:

"REUSABLE PACKAGE
DO NOT DESTROY
DO NOT OPEN PRIOR TO USE"

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The Gunner's Display is intended for use with the AN/VSG-2 on the M60A3 Tank.

6.2 Ordering data. Procurement documents will specify, as a minimum, the following:

a. Title, number, and date of this specification.
b. That quality conformance inspection lots for Group C testing (see 4.4.4) shall correspond to the quantity of Gunner's Displays to be delivered each month.
c. If there is to be a requirement to perform first article inspection and the quantity of first article Gunner's Displays required (see 4.3).

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d. Corrective action requirements for Group C inspection (see 4.4.4.4).
e. That the first article tests of table II may be performed concurrently.
f. Disposition of first article Gunner's Displays. First article Gunner's Displays shall be returned to satisfactory operation and submitted for inspection and acceptance as part of a Quality Conformance Gunner's Display lot.
g. Level A, level B or commercial preservation and packing (see section 5).
h. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
i. If the gunner's display is to be subjected to environmental testing as part of the AN/VSG-2 system level test at a sampling rate equal to or greater than that contained herein, then the environmental tests of this specification may be waived.
j. MIL-STD-810B shall be used for environmental tests of 4.8.

6.3 Definitions. For the purpose of this specification, the following definitions apply.

6.3.1 Damage. Damage is defined as degradation in specified performance, breakage, loosening, shift, failure of any finish, hardware, connection or component, evidence of condensation or moisture within the system, and deterioration in quality, or failure to operate in accordance with the requirements of this specification.

6.3.2 "g". g is defined as an acceleration or deceleration of 9.8 meters per second per second.

6.3.3 Ambient conditions. Ambient conditions are defined as follows:

Temperature	23°C, ±10°C (73°F, ±18°F)
Relative humidity	50 percent, ±30 percent
Atmospheric pressure	725, +50 mm of mercury -75 mm of mercury (28.5, +2.0 inches of mercury -3.0 inches of mercury)

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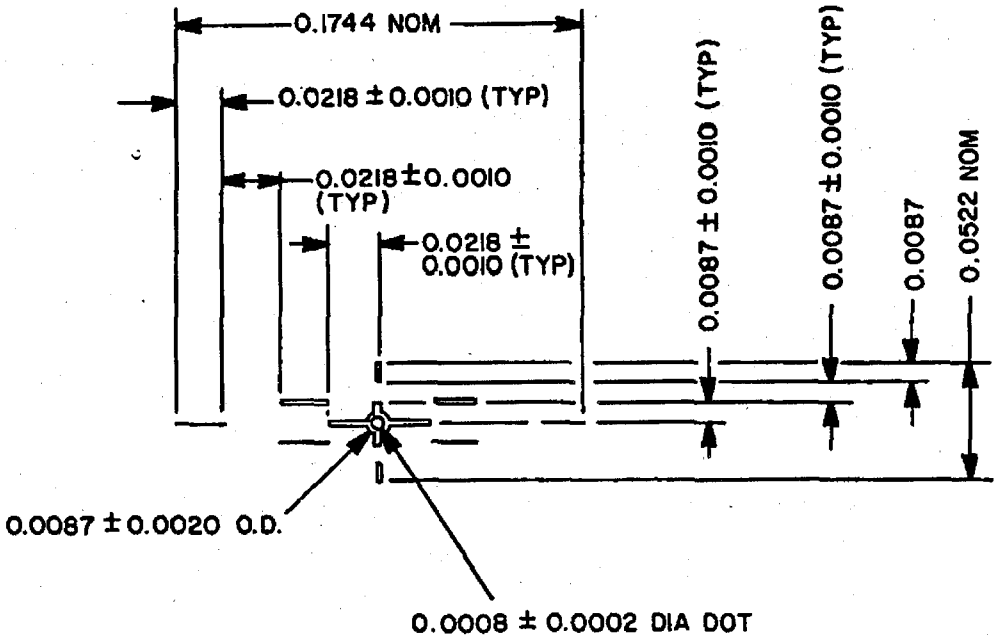


FIGURE 1. RETICLE PATTERN.

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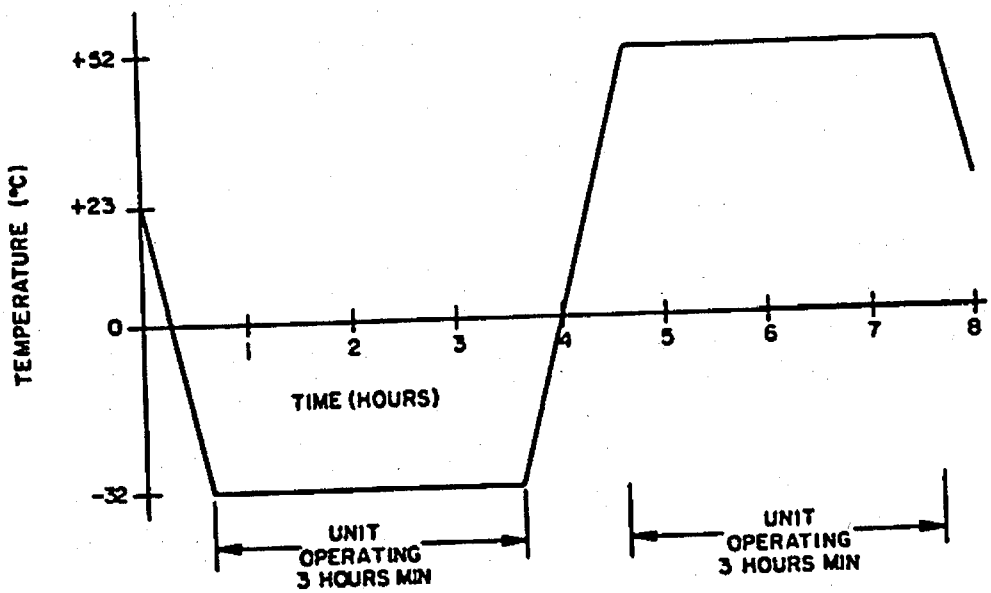


FIGURE 2. ONE PRE-CONDITIONING CYCLE.

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